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Investigating features that play a role in predicting gifted student engagement using machine learning: Video log and self-report data

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Abstract

This study aims to develop a predictive model for predicting gifted students' engagement levels and to investigate the features that are important in such predictions. Features reflecting students' emotions, social-emotional learning skills, learning approaches and video-watching behaviours were used in the prediction models. The study group consisted of 36 gifted students between the ages of 12 and 14 who attended an information technologies course, where students engaged with Arduino-based learning tasks. Data related to only one task were analysed. Prediction models were developed using different machine learning algorithms, and information gain scores were used to investigate important features in the prediction models. The results show that all prediction models achieved a higher classification accuracy than the base model. The highest classification accuracy (83%) was achieved with the Support Vector Machine (SVM) algorithm. Students' self-reported emotions while performing the task were found to be the most important feature in predicting their level of engagement. Other important features in predicting engagement level were the students' social-emotional learning skills, deep and surface learning approaches and their video-watching behaviours. As a result, it was found that the engagement level of gifted students could be predicted with high accuracy. These results can be used to predict students' engagement levels and to develop interventions for low-engaged students. The obtained results are discussed within the scope of emotion-aware learning design and social-emotional learning.

Keywords Gifted student · Emotion · Social-emotional learning · Learning approaches · Engagement · Machine learning · Video analytics

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1 Introduction

Social, economic and technological changes have led to discussions about the meaning of academic achievement and raised the question of how engagement in learning environments can be achieved in a multidimensional way. Student engagement is defined as the effort and time a student spends while achieving learning goals (Kuh, 2009). Learning environment, curriculum/activity, and the way educational technology is used are some of the factors that influence student engagement in the education settings (Bond et al., 2020; Bond & Bedenlier, 2019). Engagement has multiple dimensions: cognitive, emotional, and behavioral. Each of the cognitive, emotional, and behavioral dimensions of engagement comprises a range of indicators. For example, cognitive engagement relates to deep learning strategies, setting learning goals, and self-regulation; emotional engagement relates to interactions, reactions to the learning environment, visible expressions, and students' emotions; and behavioural engagement relates to homework completion, participation, time spent, on/off-task behavior, and action/initiation (Bond et al., 2020; Bond & Bedenlier, 2019; Henrie et al., 2015).

It is noteworthy that studies on student engagement in the literature focus on the relationships between learning, academic achievement and emotion (Alpaslan & Ulubey, 2021; Archambault et al., 2022; Camacho-Morles et al., 2021; Timonen et al., 2022; Tseng, 2021). In addition, the relationship between student engagement and various variables, such as social-emotional learning (SEL) (Greenberg et al., 2017; Rosen et al., 2022; Santos et al., 2023), academic efficacy, self-efficacy, self-esteem and self-regulation (Merhi et al., 2018; Pellas, 2014; Wang et al., 2020) have also been studied.

Online learning opportunities, such as video-based online learning, are diversifying and becoming progressively more enriching. The literature shows that the use of videos in online learning increases students' engagement and contributes to student learning (Bonafini, 2017; Chen et al., 2020; Ding et al., 2020; Hajhashemi et al., 2017; Kim et al., 2021; Lackmann et al., 2021; Mitra et al., 2010; Tseng, 2021). It has also been found that supporting teaching processes with video provides stronger learning (Noetel et al., 2021). However, video plays a role in student engagement in two ways: first are factors related to the video, such as video type, interaction with the video, video platform, video viewing intensity and video duration, and second are factors related to the individual, such as previous experiences, cognitive load, student emotions, teacher annotations and self-regulated learning strategies (Brame, 2016; Chen et al., 2020; Ding et al., 2020; Tseng, 2021). For example, students' emotions of enjoyment, excitement, annoyance and boredom in an online course are significant predictors of video engagement, and engagement is a predictor of self-perceived achievement (Ding et al., 2020). In addition, it was found that teacher annotations increased the cognitive and behavioural engagement of the students while watching the video, but did not increase their emotional engagement (Tseng, 2021).

Previous studies measured engagement in various ways using scale, facial expressions, gestures, eye movement, machine learning approach or multimodal

data (Bosch, 2016; Dewan et al., 2019; Whitehill et al., 2014). It can be said that self-report studies collect data using questionnaires, scales and interviews, while behavioural studies analyse students' facial expressions, video-watching behaviours and eye movements. Thus, there are various classifications related to detecting student engagement in the literature (Bosch, 2016; Dewan et al., 2019). In a review study, a taxonomy was proposed by dividing existing methods into three main categories (manual, semi-automatic and automatic) to detect student engagement in the context of online learning (Dewan et al., 2019). According to this taxonomy, the manual category refers to the self-reporting and observational checklist methods, the semi-automatic category refers to engagement tracing, and the automatic category refers to log-file analysis, sensor data analysis and computer vision-based methods (facial expression, eye movement, and gesture and posture) (Dewan et al., 2019). In another study, two classifications were used to detect participation: emotional and cognitive state detection (using facial features, skin conductance sensors, self-reporting), and behavioural engagement detection (using log file, machine predictions) (Bosch, 2016). Indeed, an alternative way to understand student engagement in video-based online learning is to measure students' video-watching behaviours (Guo et al., 2014). It has been stated that behavioural engagement means participation in learning; as this is related to the fact that learners have completed their video-watching behaviours, it can be measured by stopping the video (Li & Baker, 2016; Liu et al., 2018). In addition, behavioural engagement can be measured through playing, pausing and rewinding videos during learning activities (Goggins & Xing, 2016; Tseng, 2021). Cognitive engagement requires the use of self-regulation strategies, refers to the understanding of learning and can be measured by pausing and seeking the video (Li & Baker, 2016; Liu et al., 2018). While pausing in video-watching behaviour requires a higher level of cognitive interaction, seeking allows learners to jump to any part of the video while remaining in control (Liu et al., 2018; Li & Baker, 2016; Sinha et al., 2014). Emotional engagement is related to learners' emotional reactions and love of learning (Fredricks et al., 2004; Liu et al., 2018). For example, lack of interest in the content during video watching might be related to changing video speed and rewinding the video; as such, it can be measured by the number of speed changes made during the video-watching sessions (Liu et al., 2018). Although the different dimensions of engagement are measured through video-watching behaviours, Yoon et al. (2021) argued that behavioural engagement can be measured with behavioural data rather than emotional and cognitive engagement. Some studies have suggested that combining different modalities—such as observational or behavioural data on the individual's externality—with self-report data on an individual's internality can help improve the accuracy of engagement detection (Aslan et al. 2014; Bosch, 2016; Psaltis et al. 2018; Whitehill et al., 2014). Therefore, both behavioural data and self-report data were considered in this study.

It is noteworthy that in the studies in the literature that describe the relationship between engagement and various variables, the research findings on the relationship between engagement and learning approach do not support each other. For example, studies have revealed a positive and weak correlation between engagement and deep

motivation (Raleiras et al., 2021), that student engagement has a significant effect on the student's deep learning approach (Bevan et al., 2014; Reynell van der Ross et al., 2022) and that there is a positive correlation between the deep learning approach and student engagement (Teoh et al., 2014). In addition to differences in the findings, there are also differences in the treatment of engagement as a dependent and independent variable. In a study in which the video-watching behaviours of students according to their learning approaches were analysed using video analytics, it was found that, while watching the video students with a surface approach performed a statistically significant forward search compared to students using a deep learning approach, paused and played the video more randomly, watched the video more linearly and interacted less with the video (Akçapınar & Bayazıt, 2018).

The literature states that emotions affect students' engagement, learning and academic achievement (Pekrun & Linnenbrink-Garcia, 2012): positive emotions predict student engagement positively and at a high level (Yavuz, 2023; D'Errico et al., 2016), while negative emotions are correlated with low student engagement (Chevrier et al., 2019) in learning environments. Positive emotions experienced in learning environments can encourage students' perseverance and learning efforts (Pekrun et al., 2002) and interest them more in learning activities (Reschly et al., 2008), while negative emotions can limit the flexibility of using cognitive resources in learning activities, as students pay more attention to the threats around them (Derakshan et al., 2009). For these reasons, it is claimed that positive emotions experienced in learning environments tend to increase student engagement, while negative emotions can prevent student engagement (Putwain et al., 2013; Zhen et al., 2017).

It has been determined that studies on student engagement with gifted students are limited (Landis & Reschly, 2013; McCormick & Plucker, 2013; Fig. 1). However,

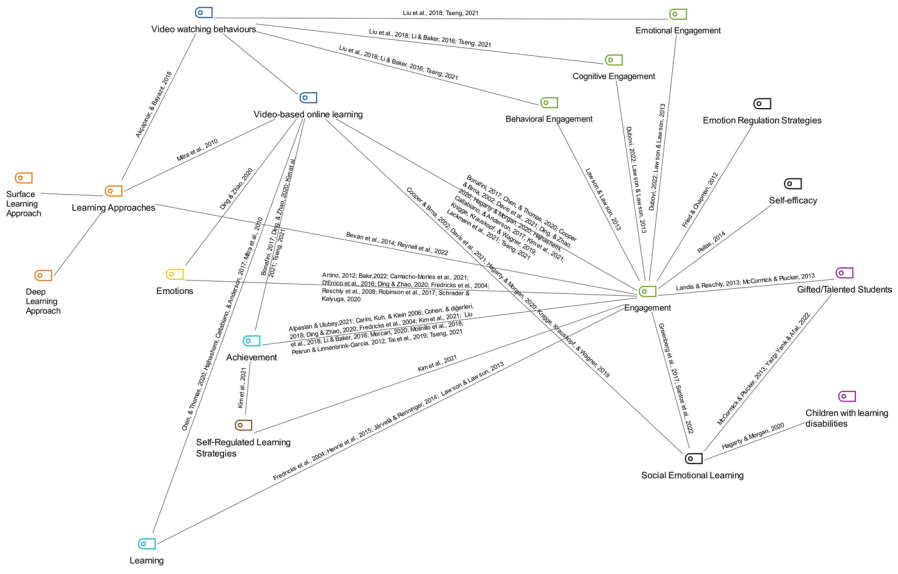


Fig. 1 Nomological network map of related literature

the engagement of gifted students is a fundamental construct for understanding, predicting and preventing dropout behaviour (Landis & Reschly, 2013). As stated in the literature, it is thought that gifted students will be successful without special services (Moon, 2009). However, gifted students who show advanced cognitive development but lag behind in emotional development may find that their abilities and strengths become a problem (Majid & Alias, 2010) and they may face having to drop out of school (Landis & Reschly, 2013). Although gifted students have high potential in terms of cognitive development and academic performance at school, they may experience social and emotional difficulties. Therefore, they should be valued and respected in the teaching–learning process (Renzulli, 2005; Pfeiffer, 2017; Ribeiro Piske & Stoltz, 2021), being taught to strengthen their abilities and receiving love and support (Casper, 2014). It is important to identify the learning differences, cognitive and emotional characteristics, as well as the academic, social, and emotional needs of gifted students and teaching strategies should be developed to enhance their learning (NAGC, 2010). It is claimed that classes in which SEL skills are taught and practiced creates such an environment (Norris, 2003). In fact, studies have shown that SEL competencies support students' well-being, encourage more positive and fewer negative social behaviours and interactions (Guerra & Bradshaw, 2008; McKown et al., 2009), improve academic achievement (Durlak et al., 2011; Weissberg et al., 2015; Zins et al., 2004) and increase high school graduation rates and university participation (Taylor et al., 2017). In addition, the literature states that SEL is positively correlated with student engagement or an important variable that predicts engagement; therefore, SEL programs should be given importance to increase engagement (Greenberg et al., 2017; Santos et al., 2023; Yang et al., 2018). In the literature, on the one hand, peer and teacher support, interaction, and collaboration in face-to-face instructional practices (Elmi, 2020), and on the other hand, videos in online education contributes to the development of students' SEL and engages them (Cooper & Brna, 2002; Davis et al., 2021; Hagarty & Morgan, 2020; Knigge et al., 2019). In the case of ICT, for example, if videos are carefully planned, designed, and integrated into classroom practice can support SEL, engagement, relationships (teacher-student and/or student–student), and motivation (Cooper & Brna, 2002). It is stated that video-based online programs provided by Moodle and Blackboard are more engaging for students in their learning experiences and contribute to the development of their SEL (Davis et al., 2021; Knigge et al., 2019). A systematic review study revealed that videos can be used as an effective technique in learning-teaching processes, especially for the development of SEL skills of children with learning disabilities (Hagarty & Morgan, 2020). It is also stated that gifted students respond favorably to self-directed projects that can add complexity to their studies compared to traditional teacher-led lectures (Maniatis et al., 2016). In this regard, it can be argued that the online environment and videos may also hold significance for gifted students' SEL skills and engagement in the learning process. The videos can be also used to encourage gifted students to adapt their learning approach according to their own preferences.

Within the scope of this study, a nomological network was created to see the study intensities and requirements between the variables/constructs in the related literature (Fig. 1). Nomological network was initially introduced by Cronbach and

Meehl (1955) to address construct validity, that is, to find and establish the meanings of relationships between constructs. These constructs and relationships can then be displayed as a map. The nomological map allows us to understand the relationship between constructs that become evident in a specific theoretical/conceptual framework, with a holistic approach (Rippee, 2020). Four guidelines have been identified in the development of a nomological network (Cronbach & Meehl, 1955):

1. Specification: Define the relationships among the model constructs based on theory and prior research.
2. Investigation: Conduct a comprehensive literature review using data collection from multiple sources and qualitative analysis.
3. Interpretation: Examine the entire network as a whole.
4. Replication: Conceptual effects can be multiplied.

In this study, a theoretical/conceptual framework of engagement's relationship with other constructs is presented with the nomological network map created as a result of comprehensive literature review. Thus, it can be more easily understood from a more holistic perspective which gaps in the literature this study fills.

As seen in this nomological network, the most studied subject in studies examining video-based online learning and video-watching behaviours is engagement, while emotion, self-regulation strategies and learning approaches are more limited; on the other hand, while the most studied subjects of engagement are academic achievement and emotion, it is noteworthy that studies on self-regulation strategies, social-emotional learning skills and emotion regulation strategies are limited, while learning approaches have not yet been studied. There are studies in the related literature on how emotions experienced in learning environments predict student engagement. These studies investigate the relationship between SEL skills and engagement or learning approaches and engagement, how students' video-watching behaviours differ according to different learning approaches, and determine the relationship between video-watching behaviours and student engagement. However, it has been determined that there is no study that addresses these variables in a holistic way.

At this point, it can be argued that examining how engagement is related to which variables among gifted students may contribute to the learning-teaching processes of these students. From this point of view, this study aims to develop a model for predicting gifted students' engagement levels and to investigate the features that are important in such predictions. In the prediction models, features reflecting students' emotions, social-emotional learning skills, learning approaches and video-watching behaviours were used.

In this study, our aim is to determine the significant features in predicting gifted student engagement and to develop a data-driven prediction model that can be used as an alternative to self-report measures. Therefore, the research questions in our study were structured as follows.

RQ1: What are the important features based on behavioural data and self-report in predicting students' engagement levels?

RQ2: To what extent can students' engagement levels be predicted when behavioural data and self-report features are used together?

2 Method

2.1 Study group

The study group of this research consisted of 36 gifted students between the ages of 12–14 who were diagnosed in the field of intellectual ability and were attending the information technologies course at a science and art centre in the spring semester of the 2021–2022 academic year. All secondary school students who chose the IT program at The Science and Art Centre and volunteered where the researcher worked as an IT teacher, were included in this study. Therefore, it can be said that a convenience sampling method was used in this study. Science and art centres are special education institutions whose purpose is to provide educational support to students who attend formal education institutions and are diagnosed as gifted in the fields of general mental ability, visual arts or music talent to develop their talents and enable them to use their capacities at the highest level (MEB, 2019). In the process of admitting students to these institutions, students attending the 1st, 2nd and 3rd grades of primary school who are deemed to show advanced differences from their peers in the areas of general intellectual ability, visual arts and musical talent are nominated by their classroom teachers. These students take part in a group screening application with a tablet computer, and students who pass the specified score threshold are taken for individual evaluation in their talent areas by The Ministry of National Education. Students who pass the evaluation phase are eligible to enrol in the science and art centre.

Twenty-four of the students in the study group were male and 12 were female, 18 were in sixth grade, 14 were in seventh grade and 4 were in eighth grade. Before starting the data collection process, ethics commission consent from Hacettepe University, Research Application Consent from The Ministry of National Education (22.03.2022 / 46,193,093) and parent permission were obtained. Students were informed and obtained consent forms about voluntary participation.

2.2 Data collection tools

To develop prediction models that predict students' level of engagement, self-report and behavioural data sources were used together. Self-reported data were collected using the Student Engagement Scale in Online Learning Environments (Sun and Rueda 2012), the Geneva Emotion Wheel (Scherer et al., 2013), the Social Emotional Learning Skills Scale (Kabakçı & Korkut Owen, 2010) and the Revised Two Factor Study Process Questionnaire (Biggs et al., 2001). Behavioural data were collected using a special video player that can record students' video interactions.

2.2.1 The student engagement scale in online learning environments

The Student Engagement Scale, originally developed by Sun and Rueda (2012) and adapted into Turkish by Ergün and Usluel (2015), has 19 5-point Likert-type items and consists of three sub-dimensions: emotional, cognitive and behavioural engagement. The Cronbach's alpha reliability coefficient for the three sub-dimensions of the scale were as follows: 0.846 for emotional, 0.833 for cognitive and 0.707 for behavioural. In addition, the Cronbach's alpha internal consistency coefficient of the scale was 0.898. The authors validated the scale with 230 gifted middle school students. As a result of the confirmatory factor analysis, two sub-dimensions of the scale (emotional, behavioural), which were compatible with the original, were validated with 230 gifted middle school students. The factor load of one of the cognitive sub-dimensions was found to be below 0.50 and was removed from the scale. The scale was applied to 18 items for this study.

2.2.2 The geneva emotion wheel

The Geneva Emotion Wheel (GEW) is a simple tool (see Fig. 2) which helps participants to self-report their emotional states (Scherer et al., 2013). The locations of the emotions in the wheel are organised in a theoretically validated two-dimensional system consisting of valence and strength dimensions. The GEW is organised in a wheel-like format consisting of 20 words representing the emotions experienced and five circles representing the intensity of each emotion (Scherer et al., 2013).

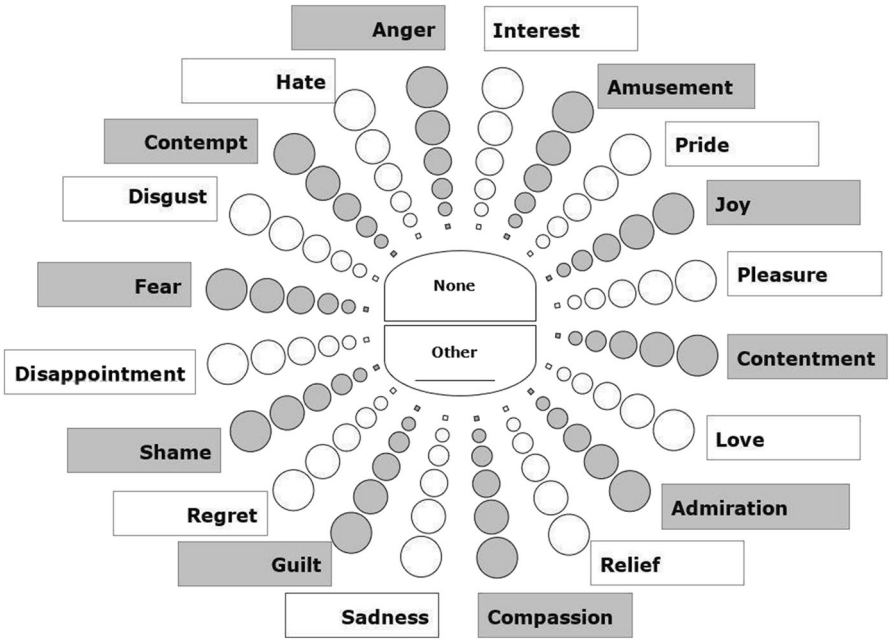


Fig. 2 The Geneva Emotion Wheel (Scherer et al, 2013)

Students can write emotions they experience other than those emotions written on the wheel in the "Other" section at the centre of the wheel.

2.2.3 The social emotional learning skills scale

The Social Emotional Learning (SEL) Skills Scale was developed by Kabakçı and Korkut Owen (2010) to measure the SEL skills of primary school students. The scale has 40 4-point Likert-type items and consists of four sub-dimensions: problem solving (PS), communication (C), self-worth (SW) and coping with stress (CS). Validation of the scale with gifted middle school students was conducted by the authors with 230 gifted middle school students. Exploratory and confirmatory factor analyses were conducted, and construct validity was ensured. The Cronbach's alpha values of each sub-dimension were calculated as 0.80 for PS, 0.79 for SW, 0.69 for CS and 0.70 for C, and were found to be at acceptable levels. The Cronbach's alpha internal consistency coefficient of the scale was determined to be 0.88. The 4-factor structure of the scale was confirmed for gifted middle school students. A low score on the scale indicates inadequacy in terms of SEL skills, while a high score indicates competence in terms of SEL skills.

2.2.4 Revised two factor study process questionnaire (R-SPQ-2F)

The Revised Two Factor Study Process Questionnaire (R-SPQ-2F) was published in 2001 (Biggs et al., 2001) and adapted into Turkish by Tetik et al. (2009). Validation of the Turkish scale with gifted middle school students was conducted by the authors with 230 gifted middle school students. The scale consists of 20 5-point Likert-type items and two subscales: Deep Approach (DA) and Surface Approach (SA). In the reliability analysis of the scale, the Cronbach's alpha values of each sub-dimension were calculated respectively as 0.835 for the DA and 0.741 for the SA and found to be at acceptable levels. The total scores obtained from the deep and surface subscales were calculated separately.

2.2.5 Video player

The video-watching behaviours of the students were collected with a video player embedded in the Moodle learning management system. This video player, shown in Fig. 3, records the Load, Play, Pause and Seek activities performed by the students. In addition, the video player triggers an automatic event at five-second intervals, which shows that the video continues to play even if the user is not performing any activity. All events were logged into the database as a click-stream, including students' user IDs and timestamp. The video player helped to obtain a reflection of the video-watching behaviours of the students. In the data pre-processing step, video-related features were extracted for each student by using the data recorded in the database.

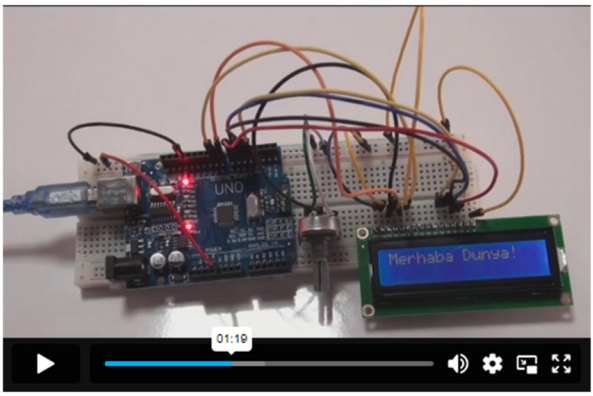


Fig. 3 Video player

2.3 Data collection process

This application contains only the online two-week part of the Information Technologies (IT) course, which covers the 6-week program conducted in a hybrid way in the spring semester of the 2021–2022 academic year. The 4 weeks of the program were carried out with face-to-face lectures by the teachers, and the last 2 weeks with the videos watched over the learning management system (LMS) in the classroom environment. The course lasts four hours per week. Each week of the course, students were given a real-life task, such as a smart home or smart agriculture, and asked to solve it using an Arduino microcontroller and electronic components such as sensors, LCD displays and motors. During the two-week application period, students watched four videos prepared by their course teachers through LMS. In application, students' participation in LMS was carried out in the classroom environment. During this application, two IT teachers conducted the lesson together. One of the IT teachers is also one of the researchers. The other researchers made observations during each lesson and ensured that data collection tools were completed by the students. In the first week, students were provided with usernames and passwords to access the online environment. They were then instructed to individually watch the videos and apply what they learned. Internet connectivity, video data recording, and necessary controls were conducted during this week. In addition, two videos are provided to demonstrate the use of components required for the task, the smart home project. In the second week, two videos are provided to demonstrate the use of components required for the task, the smart agriculture project. Additionally, an extra video is available for students who require additional activities. Video materials explaining the connections of the different circuit elements were given to the students. The students worked individually on a task and two ICT teachers were present in the class to assist them when they needed technical help. In the first week, taking into account the adaptation of the students to the LMS, the logs were taken over the video in the second week. Students' video interactions were recorded into the LMS's database. After completing the task, students answered the SEL Skills Scale, the R-SPQ-2F and the Student Engagement Scale. While performing the task, they were asked to self-report their emotions through the Geneva Emotion Wheel.

Considering the changeable nature of emotions, in this study the GEW was assessed separately for each task; it was completed by students three times: before the task was given, during the task (watching the video), and after the task was completed (Fig. 4).

2.4 Data analysis

In this study, the relationship between the two research questions is that one is related to development of the prediction model that will perform the best, while the other involves determining the features that are crucial in this prediction process. This enables us to understand which features the prediction model actually classifies students based on, in other words, which features are more decisive in determining whether students have low or high engagement.

Research on the development of prediction models is widespread in the field of predictive modelling in teaching and learning. Particularly, in Educational Data Mining (EDM), Learning Analytics (LA), and Artificial Intelligence in Education (AIED) domains, researchers conduct studies on the development and implementation of such models. One of the crucial data preprocessing steps in creating prediction models is feature selection. In this stage, features with high distinctiveness on the target feature are identified from a large number of candidate features, and the prediction model is constructed using these features. This not only reduces the time and computational power required for training the model but also enhances the interpretability of the model. Another application of feature selection is determining the significant features in the prediction model, especially in the field of education (Romero et al., 2013b). The common approach in the development of prediction models based on machine learning involves comparing different algorithms and methods to determine the algorithm that performs best on the specific dataset (Gray & Perkins, 2019; Raj & VG, 2022; Romero et al., 2013a; Wang et al., 2021).

Prediction models were created using Orange data mining software according to the process given in Fig. 5. First, the extracted features were read from a file and related columns were filtered.



Fig. 4 Students performing the task in the classroom

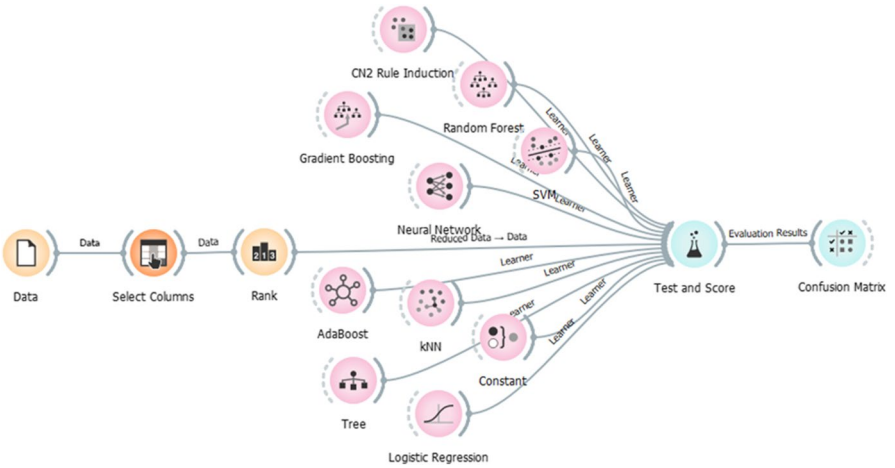


Fig. 5 Data analysis process

The features used in the prediction models and their descriptions are presented in Table 1. These features are: the emotional states of the students during the task, the total scores they obtained from the SEL Skills Scale, the total scores they obtained from the surface approach and deep approach dimensions of the R-SPQ-2F questionnaire, and the features related to their video-watching behaviours.

The total scores of the students obtained from the Student Engagement Scale in Online Learning Environments were used as the prediction variable. We formulated an engagement prediction task as a binary classification problem. As stated by the authors (Ergün & Usluel, 2015) of the scale used, a high score obtained from the scale indicates a high level of engagement to the online learning environment; a low score means that the level of engagement is low. Therefore, students' scale scores were discretized into two categories: low and high using the median split (Iacobucci et al., 2015) method. The median value of 59 was taken as the cut-off point. In other words, students with an engagement score of 59 and above (n = 20) were labelled

Table 1 Features used in prediction models and their descriptions

Feature	Description
emotional_state	Self-reported emotional state of the student while engaging the task
sel_score	The total score obtained from the Social Emotional Learning Skills Scale
surface_approach_score	The surface approach score obtained from the R-SPQ-2F
deep_approach_score	The deep approach score obtained from the R-SPQ-2F
video_duration	Video watching time of the student (seconds)
video_pause_and_play	Number of times student paused and played the video
video_pct	Video watching percentage of the student
video_seek	Number of seeks made by the student on the video timeline

a high engagement group, while students below 59 (n = 16) were labelled as a low engagement group.

In the second stage, important features were selected using the Rank widget. The information gain scores of the features were used in ranking, and prediction models were then created with the top 5 ranked features. Since the algorithms can produce different results in different data sets, prediction models were created with different algorithms and their performances were compared. Algorithms widely used in educational studies were selected for this purpose: Support Vector Machines (SVM), AdaBoost, CN2 rule inducer, Gradient Boosting, k Nearest Neighbour (kNN), Logistic Regression, Neural Network, Random Forest and Decision Tree.

Area Under the ROC Curve (AUC), Classification Accuracy (CA), F Measure (F1), Precision and Recall were used to evaluate the performance of the algorithms. The 5 k-cross validation method was used to calculate performance metrics. The AUC, F1, Precision and Recall metrics were calculated separately for each case (high and low engagement), as positive cases and the average values were reported. Algorithm performances were also compared with the Constant model, which represents majority class classification.

3 Results

Box plots of the numeric features for each group are shown in Fig. 6. When the features reflecting the video-watching behaviours of the students were examined, it was seen that, except for video_duration, they have a similar distribution. As for video watching time, students in the high engagement group spent more time

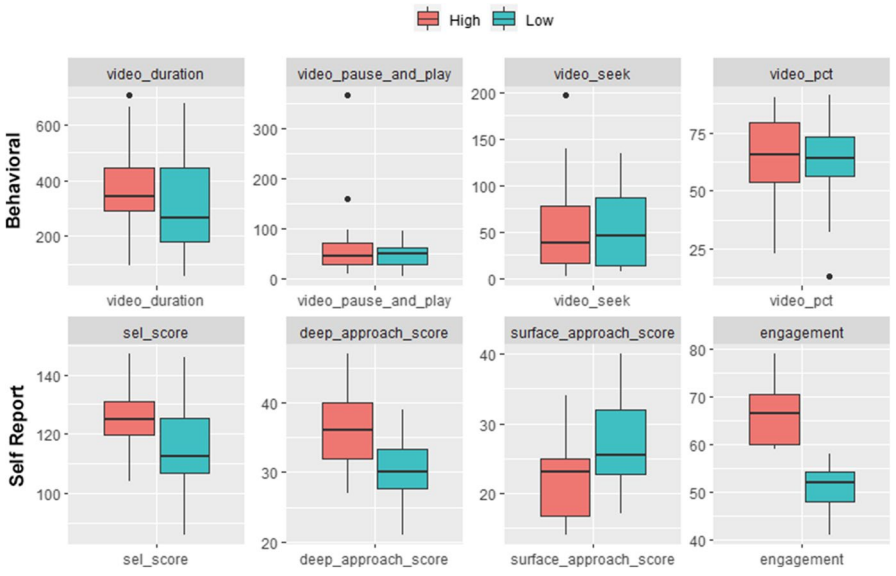


Fig. 6 Box plots of numerical features

watching videos than did the students in the low engagement group. When the distribution of the self-report features was examined, it was seen that the students in the high engagement group had higher values, except for the surface approach scores, which were higher in the low engagement group.

Students' self-report emotions before, during and after the task, according to their low–high engagement levels, are shown in Table 2, Table 3 and Table 4. When analysing the emotions before the task in relation to high and low levels of engagement, it was found that students with high engagement group experienced the emotions of excitement (n = 7) and joy (n = 5) (Table 2). In contrast, students with low engagement reported joy (7) and interest (4) as the most common emotions (Table 2). One student from each engagement level expressed anxiety before the task was given.

When the emotions experienced by the students while performing the task were examined according to the high and low levels of engagement, students in the high engagement group stated that the emotions they experienced the most were excitement ($n = 9$) and joy ($n = 7$) (Table 3). The students in the low engagement group, on the other hand, stated that the emotions they experienced the most were joy (6) and interest (5).

In the high engagement group, students reported emotions of pride (7), joy (6), and relief (3) after completing their robot tasks. Conversely, students in the low engagement group expressed emotions of joy (9), relief (3), and pride (2) after finishing their tasks (Table 4).

Table 2 Students self-reported emotions before the task

	Emotions							
Engagement	joy	excitement	interest	pleasure	curiosity	sadness	anxiety	none
High (f)	5	7	3	-	3	-	1	-
Low (f)	7	1	4	-	2	-	1	1

Table 3 Students self-reported emotions while performing the task

	Emotions							
Engagement	joy	excitement	interest	pleasure	curiosity	sadness	anxiety	none
High (f)	7	9	2	2	-	-	-	-
Low (f)	6	1	5	-	1	1	1	1

Table 4 Students self-reported emotions after completing their tasks

	Emotions							
Engagement	joy	excitement	interest	pleasure	contentment	pride	relief	none
High (f)	6	2	-	2	-	7	3	-
Low (f)	9	-	-	-	1	2	3	1

According to these findings, it can be said that the emotions experienced by students are related to the process of performing learning tasks. In addition, it can be argued that their task-related thoughts, that is, their cognitive appraisal, are also effective in the formation of their emotions.

3.1 Findings related to the first research question

Within the scope of the first research question, we aimed to determine the important features in predicting students' engagement levels. For this, we used the Rank module in the Orange data mining software. Information gain scores were taken into account to rank the features according to their importance in separating low- and high-engaged students. When the features given in order according to their information gain scores in Fig. 7 were examined, the self-reported emotional state of the students during the task (emotional_state) was found to be the most important one in predicting their level of engagement. In other words, the self-reported emotional states of the students while performing the given task were found to be the most important variable in determining whether the student was classed in the low or high engagement group. Based on the information gain scores, the second important variable was the total score obtained by students on the SEL Skills Scale (sel_score). The scale has four sub-dimensions: problem-solving skills, communication skills, self-esteem skills and stress coping skills. Low scores on the scale indicate inadequate social-emotional learning (SEL) skills, while high scores indicate sufficient SEL skills. This was followed by Deep Approach scores, Surface Approach scores. Deep approach and surface approach scores represent students' approaches to learning based on their scores on the R-SPQ-2F scale. Next features are the number of pause and play actions (video_pause_and_play) while watching the video and other video-watching behaviours included the percentage of video watched (video_pct), the duration of video watched (video_duration), and video-seeking behaviour (video_seek) (Fig. 7).

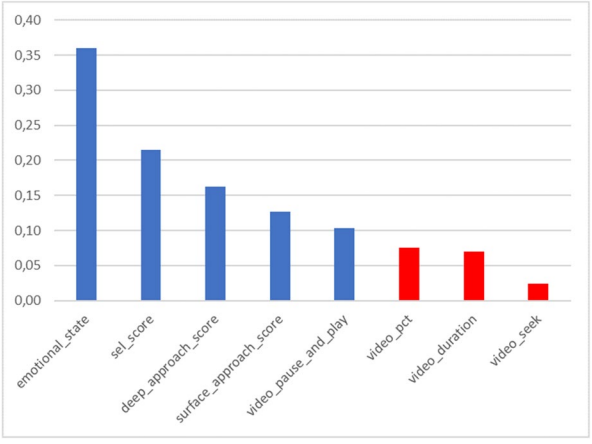


Fig. 7 Feature importance based on information gain scores

The percentage of the video they watched, the total time they spent on the video and the number of seeks they made while watching the video were less important in predicting the level of engagement of the students.

3.2 Findings related to the second research question

Within the scope of the second research question, we investigated the extent to which the students' level of engagement could be predicted when the behavioural and self-reported features were used together. Features that were determined to be important in the first research question were used as predictors when developing prediction models. Here, the five top-ranked features, which also had information gain scores above 0.1, were selected. The performance of nine different prediction models (one of which was constant) was compared. The performance metrics calculated as a result of 5 k cross-validation of the created prediction models are presented in Table 5. When the results are compared in terms of Classification Accuracy (CA), it is seen that all prediction models outperformed the base model (Constant). It is also seen that the Support Vector Machine algorithm exhibited the best performance, while the Naive Bayes algorithm exhibited the lowest performance.

When the confusion matrix regarding the results of the Support Vector Machine model given in Table 6 was examined, it was seen that the Support Vector Machine algorithm correctly predicted 17 of 20 (85%) students with a high engagement level and 13 of 16 (81%) students with a low engagement level. On the other hand, the

Table 5 Prediction model results

Model	AUC	CA	F1	Precision	Recall
SVM	0.734	0.833	0.833	0.833	0.833
Neural Network	0.825	0.806	0.802	0.811	0.806
kNN	0.805	0.778	0.772	0.787	0.778
Tree	0.750	0.750	0.751	0.752	0.750
Random Forest	0.813	0.722	0.720	0.722	0.722
Logistic Regression	0.766	0.722	0.723	0.728	0.722
Naive Bayes	0.772	0.667	0.663	0.665	0.667
CN2 rule inducer	0.659	0.583	0.568	0.576	0.583
Constant	0.475	0.556	0.397	0.309	0.556

Table 6 Confusion matrix for Support Vector Machine

		Predicted		
		High	Low	Total
Actual	High	17	3	20
	Low	3	13	16
	Total	20	16	36

model misclassified 3 students with a high engagement level and 3 students with a low engagement level.

Except for the Naive Bayes and CN2 rule inducer algorithms, all models had a classification accuracy over 70% in classifying students according to their level of engagement.

4 Discussion

In line with two research questions in this study, firstly, the features that are important in predicting the engagement levels of gifted students were investigated. Secondly, prediction models were developed to predict students' engagement levels based on these features. This article concluded with the following answers to the research questions:

4.1 RQ1: What are the important features based on behavioural data and self-report in predicting students' engagement levels?

When the features in predicting students' engagement levels were examined, the emotions that students reported while engaging in the task were found to be the most important feature in predicting their engagement levels. This finding is consistent with the findings of previous studies (Artino, 2012; Yavuz, 2023; D'Errico et al., 2016; Ding & Zhao, 2020; O'Regan, 2003).

To understand emotions, it is important to know that emotions have both universal properties and individual uniqueness, and that emotions can vary depending on context and time (Pekrun, 2014). Therefore, it can be argued that identifying tasks and situations that allow students to enjoy lessons will help students develop their capacity to experience positive emotions during the learning process (Pekrun, 2014) and their emotional development.

Students reported that they experienced mostly excitement, joy and interest while performing the task. No negative emotions were expressed by the high engagement group. Some students in the low engagement group, however, expressed negative emotions such as anxiety, while a smaller number felt sadness. It can be said that our results are consistent with studies in the literature. Thus, the literature states that positive emotions are associated with high student engagement, while negative emotions are associated with low student engagement (Reschly et al., 2008). Moreover, when students experience positive emotions in a learning environment, their engagement level increases significantly (D'Errico et al., 2016). In addition, Ding and Zhao's (2020) analysis of the relation between emotions, engagement and self-perceived achievement revealed that, while positive emotions (enjoyment and excitement) predict engagement positively, negative emotions (boredom and annoyance) predict engagement negatively in the learning environment.

It can be argued that many factors may have played a role in the students' experience of positive emotions. The personal characteristics, for example, they prefer new and difficult experiences while developing their critical thinking skills, their

creative traits are strong, they try to produce a product from the fields they are studying, they develop original ideas and they like to be successful in their work (Çitil & Ataman, 2018; NAGC, 2010; Renzulli, 2005; Siegle et al., 2014) or the learning tasks related to the chosen activities or the unique structure of this special education school where only gifted students can be studied might also play a role in their positive emotions.

It was seen that the second most important feature in predicting students' engagement levels was students' social emotional learning skills. The total score obtained by the students on the Social Emotional Learning Skills Scale was important in predicting whether they were in the low or high engagement group. The literature states that social and emotional competencies are important features that are positively related to student engagement and predict engagement; thus, SEL programs should be assigned importance to increase engagement (Greenberg et al., 2017; Santos et al., 2023; Yang et al., 2018). Furthermore, as social-emotional development is an important factor for gifted and talented students to develop their other abilities and to achieve their full potential, the social-emotional needs of these students should be supported (McCormick & Plucker, 2013; Yazgı Yanık & Afat, 2022). Finally, it has been found that the engagement of gifted and talented students can be increased by identifying and supporting their academic and social emotional needs (McCormick & Plucker, 2013).

In this study, the deep learning approach was found to be another important factor in predicting students' engagement levels. Studies suggest that learning approaches affect the quality of learning outcomes, generally stating that learners with a deep approach achieve better learning outcomes than those with a surface approach (Gijbels et al., 2005). Some studies in the literature have found statistically significant positive relationships between student engagement and a deep learning approach, while there is no relationship between engagement and surface learning (Reynell van der Ross et al., 2022). Other research suggests a positive relationship between student engagement and a student-centred teaching approach with a deep learning approach (Teoh et al., 2014). In addition, studies focusing on gifted students found that positive emotions were associated with a deep approach, while negative emotions were associated with a surface approach. These studies also explored the impact of learning approaches on engagement (Yavuz, 2023).

When considering the features reflecting the students' video-watching behaviours, pausing and playing the video was identified as the most prominent behaviour in predicting their level of engagement. Previous studies also reported that learner engagement can be measured by video-related metrics such as rewinding the video, seeking, changing the play speed, and time spent with the video (Liu et al., 2018; Li & Baker, 2016; Sinha et al., 2014; Tseng, 2021). Also, Ding and Zhao (2020) have combined self-reported emotional data with behavioural data, and one such study found that pleasure, excitement, boredom, and fatigue were all significant predictors of video engagement. In our study, video-related features were found to be less important than the self-reported features in predicting students' engagement levels. This finding may be due to the fact that the students had to watch the videos to complete the assigned learning task. All of the students completed the task, regardless of their engagement level.

4.2 RQ2. To what extent can students' engagement levels be predicted when behavioural data and self-report features are used together?

For RQ2, prediction models were developed to predict students' engagement levels based on important behavioural and self-reported features (emotional state, SEL scores, deep approach scores, surface approach scores, and the number of pause and play activities while watching the video). When the findings related to the prediction of the students' engagement levels were examined, it was seen that different machine learning algorithms achieved high accuracy in predicting the students' engagement level as low or high. All models outperformed the prediction made by the constant model, which assigned all students to the majority class without using an algorithm. The best-performing algorithm (Support Vector Machine) classified the students with 83% accuracy. The performance of algorithms in machine learning problems may vary depending on the data set and problem situation. For this reason, comparing different algorithms is a widely used approach (Gray & Perkins, 2019; Raj & VG, 2022; Romero et al., 2013a; Wang et al., 2021). Here, it is important to choose algorithms with different characteristics when developing prediction models. If the aim is to predict students' level of engagement with the greatest accuracy, the prediction model created with the Support Vector Machine algorithm, known as the black box, can be used, but if the aim is to understand the underlying relationships between features while performing the prediction, then the prediction model created with the Decision Tree algorithm (75% accuracy) can be used.

This study shows that the level of engagement of gifted students engaged in a task could be predicted. Thus, this study contributes to the limited number of studies examining student engagement with gifted students in the literature (Landis & Reschly, 2013; McCormick & Plucker, 2013). The features that are important in this prediction model were also revealed. Predicting students' engagement levels is important in terms of developing interventions for those students with low engagement levels.

5 Conclusion

This study developed models for predicting gifted students' engagement levels and to investigate the features that are important in such predictions. In the prediction models, features reflecting students' emotions, social-emotional learning skills, learning approaches and video-watching behaviours were used. Prediction models were developed using different machine learning algorithms, and information gain scores were used to investigate important features in the prediction models. All prediction models achieved a higher classification accuracy than the base model. The highest classification accuracy (83%) was achieved with the Support Vector Machine (SVM) algorithm. Students' self-reported emotions while performing the task were found to be the most important feature in predicting their level of engagement. Other important features in predicting engagement level were the students' social-emotional learning skills, deep and surface learning approaches and their video-watching behaviours.

As a result, it was found that the engagement level of gifted students could be predicted with high accuracy. These results can be used to predict students' engagement levels and to develop interventions for low-engaged students. It can also be argued that the results obtained are important in emotion-aware learning design and social-emotional learning.

6 Limitations and future research

Due to the study's limited sample size, predicting students' engagement levels was considered a binary classification problem. This situation caused students whose engagement scores were close to the cut-off value (median) to be in either the low or high group, despite having similar scores. Further studies could assign students in the middle group to a separate class and investigate the matter as a multiclass classification problem. The external validity of the prediction models can also be tested with larger sample sizes. Moreover, students' engagement levels in learning environments that consider their emotions can be studied longitudinally and in depth.

Students' video-watching behaviours might be related to learning tasks and video types. Therefore, future studies need to analyse students' video-watching behaviours and their relation with engagement in different learning settings. In addition to the behavioural and self-report data used in this study, future studies could use additional measures, such as biometrics, to provide a deeper understanding of detecting student engagement.

Since emotions are an important predictor of students' engagement level, further studies can be designed to test the effects of emotions in different instructional settings. In addition, it can be argued that, based on the fact that positive emotions are positively related to engagement, the learning situations that play a role in students' positive emotions should be investigated in depth and considered for implementation in instructional design.

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Data availability The datasets generated during and/or analysed during the current study are available from the corresponding author on reasonable request.

Declarations

Ethics approval The study complies with Ethical Standards.

Consent to participate The participants all agreed to take part in this study.

Consent for publication The publication of this study has been approved by all authors.

Conflicts of interests/Competing interests The authors declare that they have no conflicts of interests.

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